

Cheatsheet: Interactive Code with Pyodide

Running Python live in the browser

1 — Install & enable

⚠ Interactive in HTML only

Running and editing works only in **HTML output** (`format: html`), that is in websites and books. In PDF or Word the same code appears as ordinary code that cannot be run — see the Exercise Sheet cheatsheet for code that has to work everywhere.

`pyodide-interaktiv` is a rework of `coatless-quarto/pyodide` that adds real `input()`, animations, a Stop button, and AI feedback. It installs like any other extension:

Listing 1 Terminal

```
quarto add Erasmus-CTM/pyodide-interaktiv
```

Enable it in a page's YAML header:

```
---
title: "My Page"
format: html
filters:
  - Erasmus-CTM/pyodide-interaktiv
---
```

2 — Writing interactive cells

Use a fence tagged `{pyodide-python}` instead of a regular code block:

```
```{pyodide-python}
x = [1, 2, 3, 4, 5]
print(sum(x))
```
```

Students can edit this code directly in the browser and run it with the **Run code** button (or **Shift + Enter**) — nothing to install on their end.

Check it: run `quarto preview`. The cell appears with a row of buttons. On the first click on **Run code** Python loads, which takes a few seconds.

💡 Want graded exercises instead?

`py-exercise` adds code blocks with hidden automated tests; `math-exercise` checks symbolic/numeric answers with SymPy. Both are installed the same way (`quarto add Erasmus-CTM/py-exercise`, `quarto add Erasmus-CTM/math-exercise`) — live examples are on the Example Page of this tutorial.

3 — What pyodide-interactive gives you

Python runs in a **Web Worker**, that is next to the page rather than inside it. As a result:

| Feature | What it means |
|---------------------------|--|
| Real <code>input()</code> | A cell can pause and wait for typed input while the page stays usable |
| Animations | A Matplotlib <code>FuncAnimation</code> plays as an interactive player |
| Safe infinite loops | An accidental <code>while True:</code> does not lock up the page — Stop cancels it |
| AI feedback | Every cell has a Feedback button (needs to be set up first); the hints get more concrete with each click |
| Responsive page | Long calculations don't freeze the page |
| Shared variables | All cells on a page belong to one Python session. A later cell can use what an earlier one defined or imported, once that one has been run |

Try `input()` and shared variables live on the [Code page](#), and the animation on the [Visualizations page](#) of the Example Page.

Full details: [Interactive Python with Pyodide](#)